

# History of Deep Learning

Tanveer Ahmed

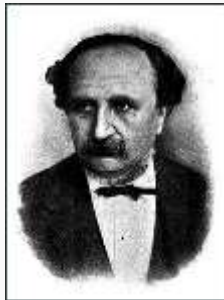
## Acknowledgements

- Most of this material is based on the article "Deep Learning in Neural Networks: An Overview" by J.Schmidhuber<sup>[1]</sup>

## Biological Neurons

## Reticular Theory

Joseph von Gerlach proposed that the nervous system is a single continuous network as opposed to a network of many discrete cells!



1871-1873

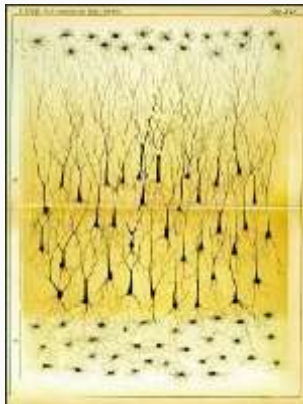


Reticular theory

## Staining Technique

Camillo Golgi discovered a chemical reaction that allowed him to examine nervous tissue in much greater detail than ever before

He was a proponent of Reticular theory.



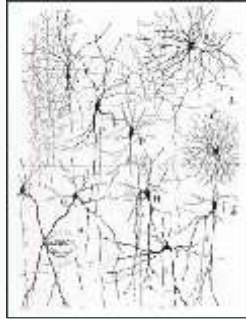
1871-1873



Reticular theory

# Neuron Doctrine

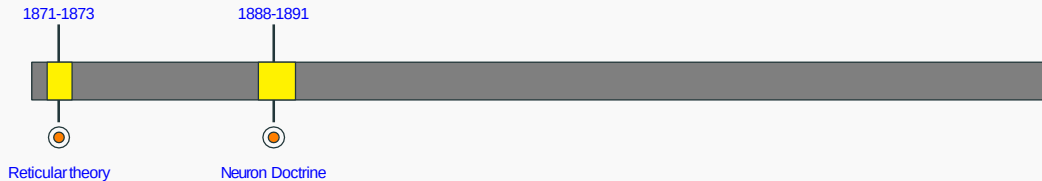
Santiago Ramón y Cajal used Golgi's technique to study the nervous system and proposed that it is actually made up of discrete individual cells forming a network (as opposed to a single continuous network)



# The Term Neuron

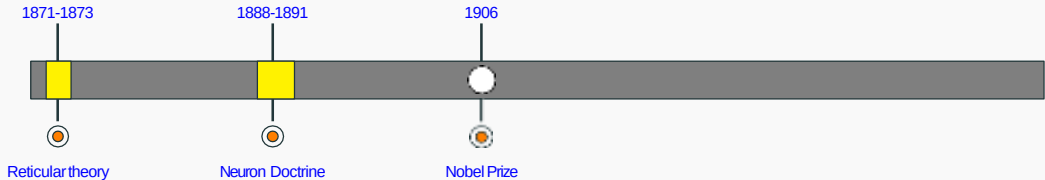
The term neuron was coined by Heinrich Wilhelm Gottfried von Waldeyer-Hartz around 1891.

He further consolidated the Neuron Doctrine.



## Nobel Prize

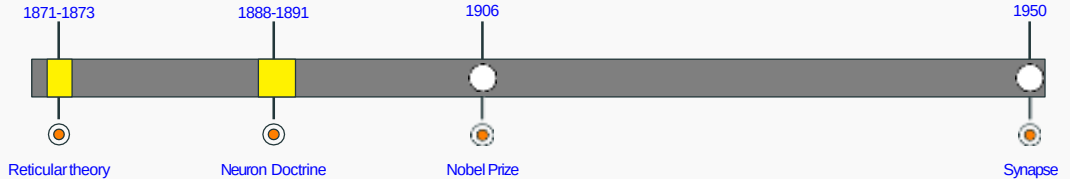
Both Golgi (reticular theory) and Cajal (neuron doctrine) were jointly awarded the 1906 Nobel Prize for Physiology or Medicine, that resulted in lasting conflicting ideas and controversies between the two scientists.



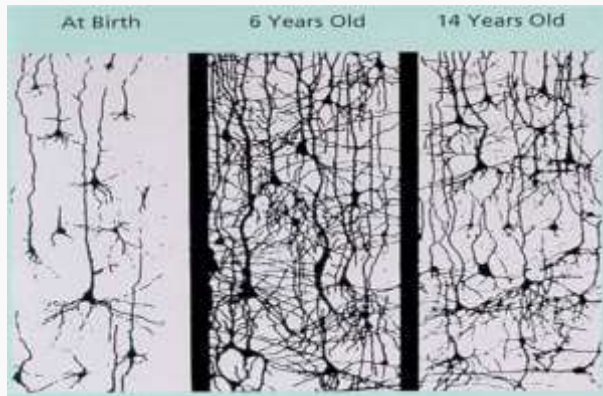
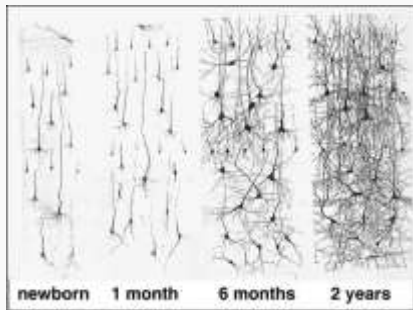


## The Final Word

In 1950s electron microscopy finally confirmed the neuron doctrine by unambiguously demonstrating that nerve cells were individual cells interconnected through synapses (a network of many individual neurons).



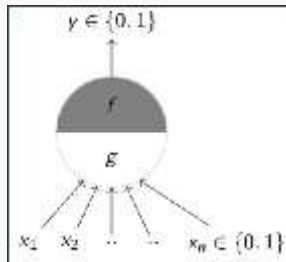
## Interesting Fact



## **From Spring to Winter of AI**

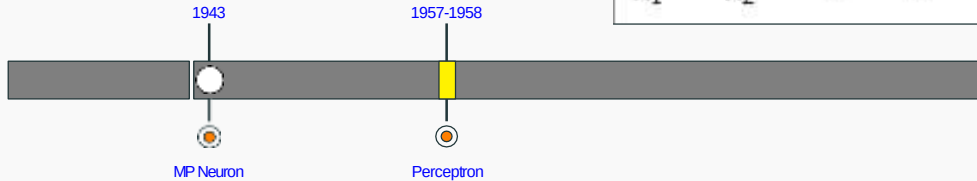
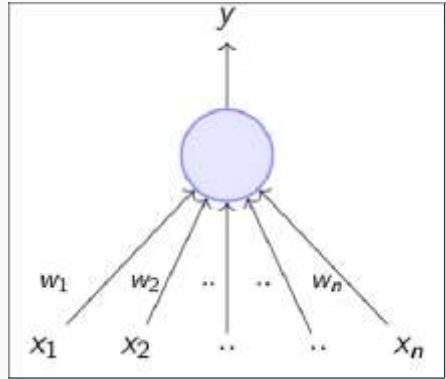
# McCulloch Pitts Neuron

McCulloch (neuroscientist) and Pitts (logician) proposed a highly simplified model of the neuron (1943) [\[2\]](#)



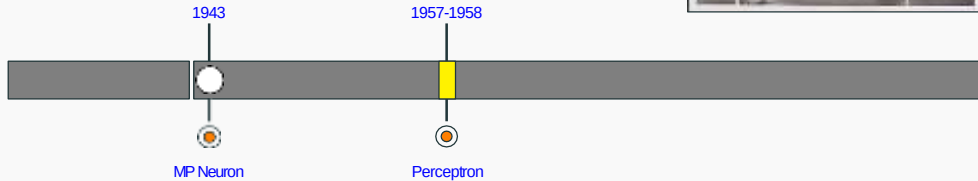
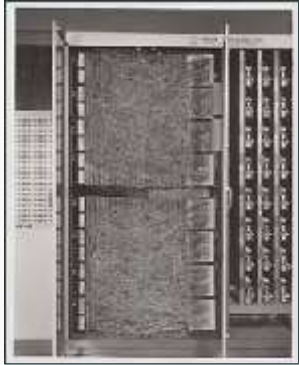
# Perceptron

“the perceptron may eventually be able to learn, make decisions, and translate languages” -Frank Rosenblatt



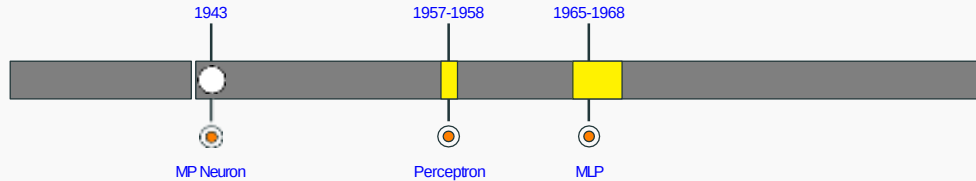
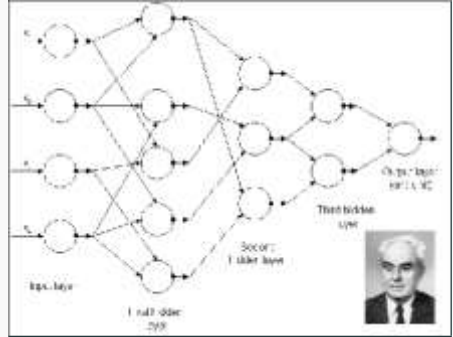
# Perceptron

“the embryo of an electronic computer that the Navy expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence.” -New York Times



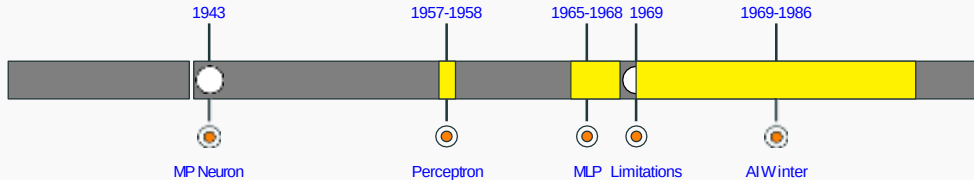
# First generation Multilayer Perceptrons

Ivakhnenko et. al. [\[3\]](#)



# AI Winter of connectionism

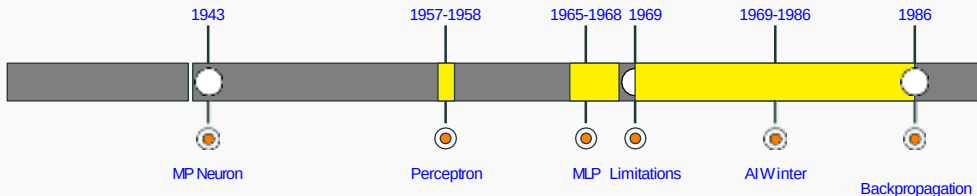
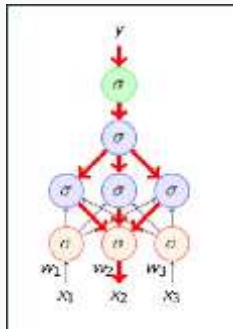
Almost lead to the abandonment of  
connectionist AI





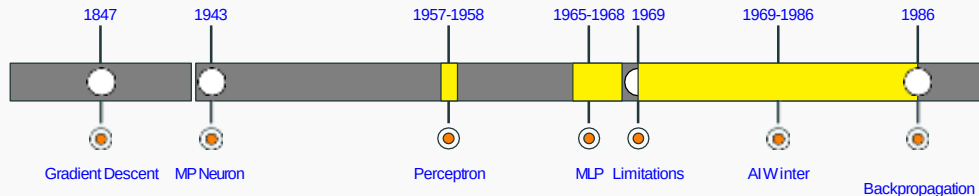
# Backpropagation

- Discovered and rediscovered several times throughout 1960's and 1970's
- Werbos(1982) [\[5\]](#) first used it in the context of artificial neural networks
- Eventually popularized by the work of Rumelhart et. al. in 1986 [\[6\]](#)



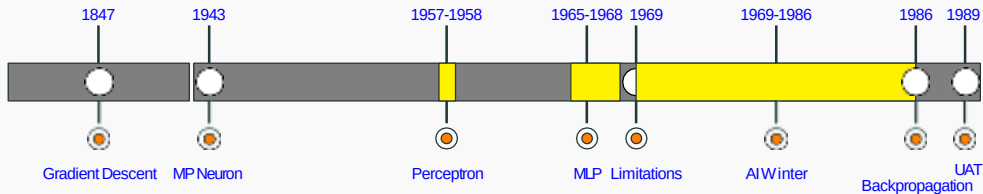
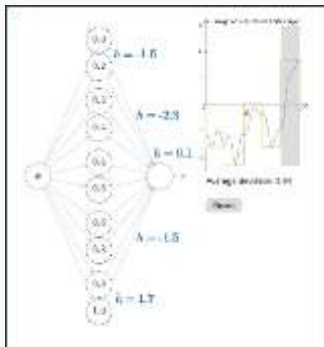
# Gradient Descent

Cauchy discovered Gradient Descent  
motivated by the need to compute the orbit  
of heavenly bodies



# Universal Approximation Theorem

A multilayered network of neurons with a single hidden layer can be used to approximate any continuous function to any desired precision [7]

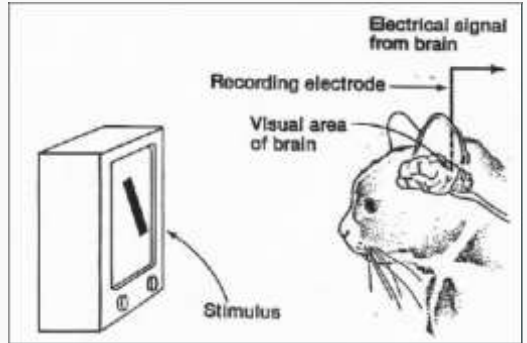


## The Deep Revival

# From Cats to Convolutional Neural Networks

# Hubel and Wiesel Experiment

Experimentally showed that each neuron has a fixed receptive field - i.e. a neuron will fire only in response to a visual stimuli in a specific region in the visual space [\[18\]](#)



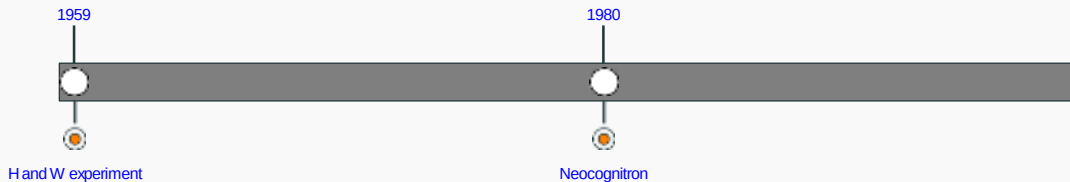
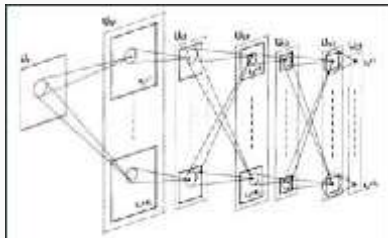
1959



Hubel and Wiesel experiment

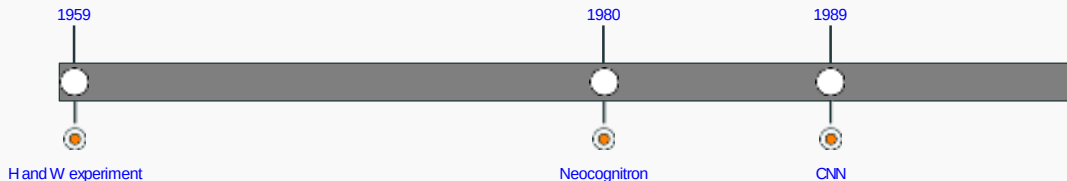
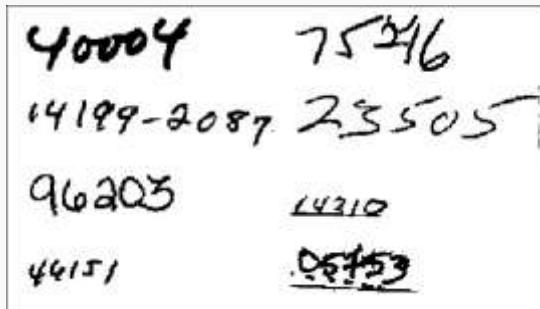
# Neocognitron

Used for Handwritten character recognition and pattern recognition (Fukushima et. al.)[\[19\]](#)



# Convolutional Neural Network

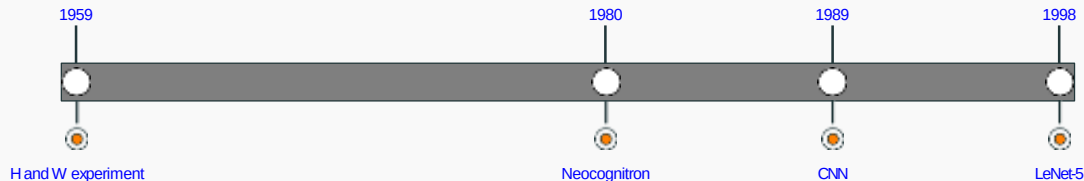
Handwriting digit recognition using  
backpropagation over a Convolutional Neural  
Network (LeCun et. al.) [\[20\]](#)





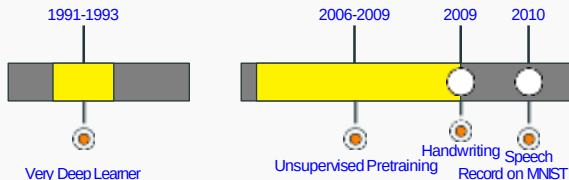
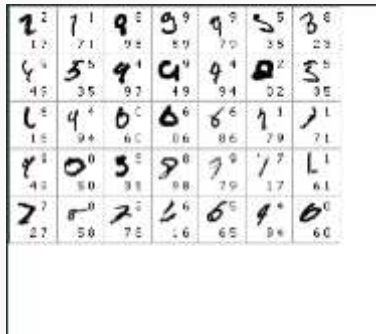
# LeNet-5

Introduced the (now famous) MNIST dataset  
(LeCun et. al.) [\[21\]](#)



## New record on MNIST

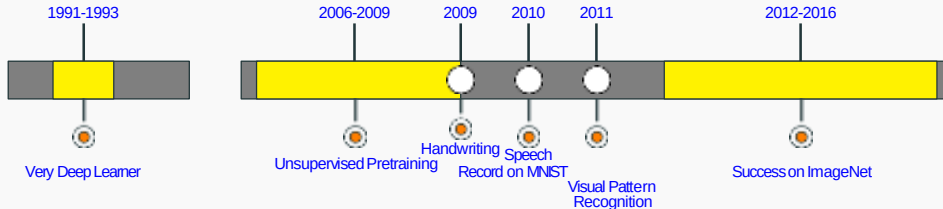
Ciresan et. al. set a new record on the MNIST dataset using good old backpropagation on GPUs (GPUs enter the scene) [\[11\]](#)



# Winning more visual recognition challenges



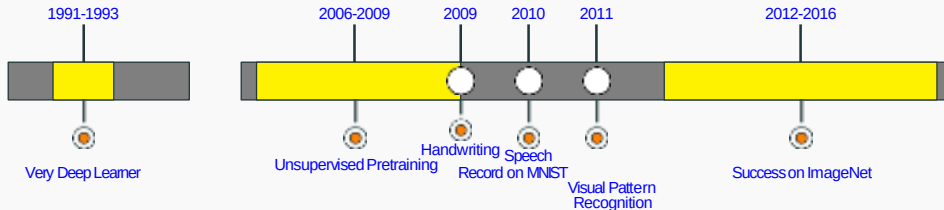
| Network                 | Error | Layers |
|-------------------------|-------|--------|
| AlexNet <sup>[13]</sup> | 16.0% | 8      |



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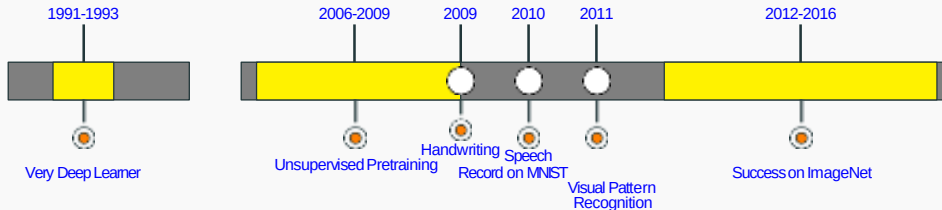
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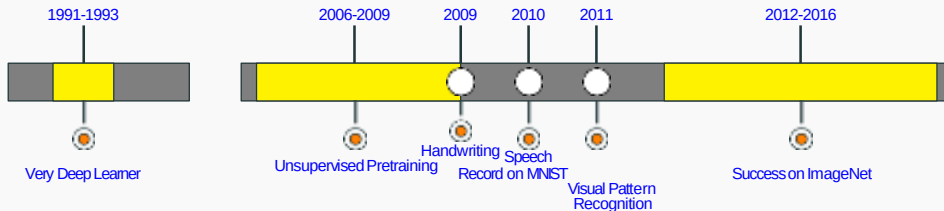
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| VGGNet <sup>[15]</sup>  | 7.3%  | 19     |



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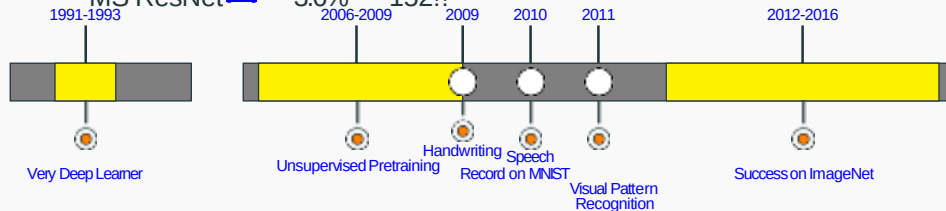
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| GoogLeNet <sup>[16]</sup> | 6.7%  | 22     |



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| MS ResNet <sup>[17]</sup> | 3.6%  | 152!!  |



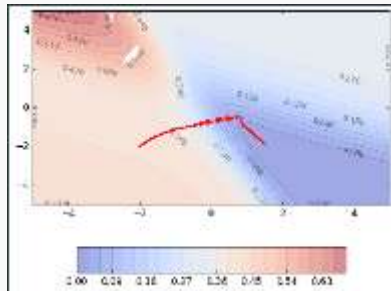
**An algorithm inspired by an experiment on cats is today used to detect cats in videos :-)**



**Faster, higher, stronger**

# Better Optimization Methods

Faster convergence, better accuracies



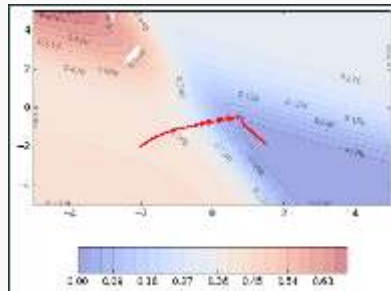
1983



Nesterov

# Better Optimization Methods

Faster convergence, better accuracies



1983



Nesterov

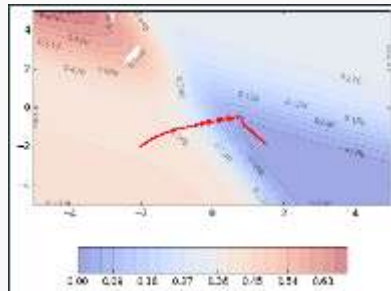
2011



Adagrad

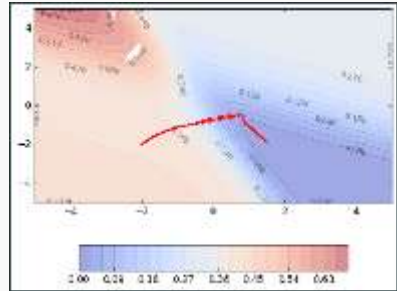
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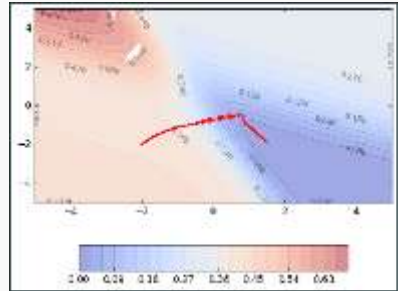
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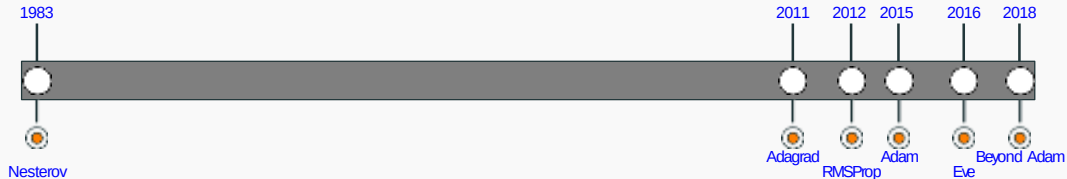
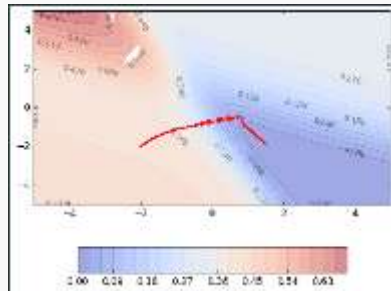
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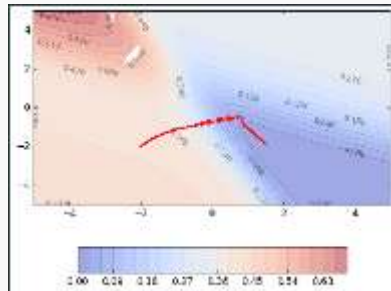
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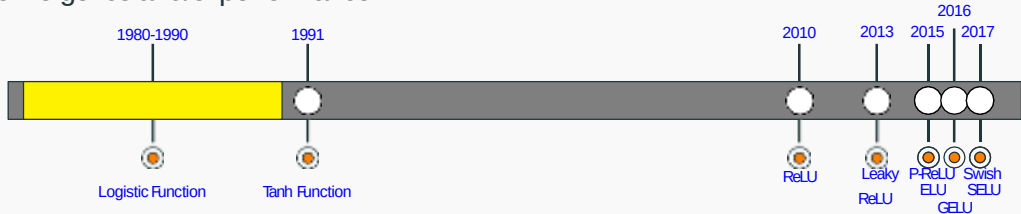
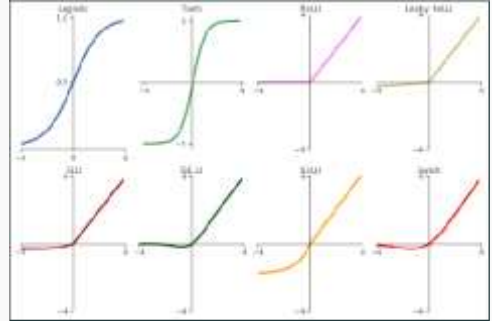


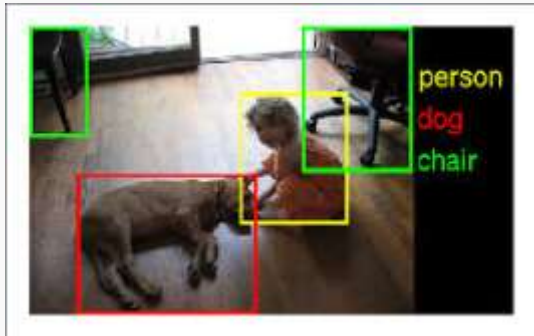


# Better Activation Functions

We have come a long way from the initial days when the logistic function was the default activation function in NNs!

Over the past few years many new functions have been proposed leading to better convergence and/or performance!





- Semantic Segmentation (Long et al, 2015) [\[59\]](#)
- Recurrent CNNs (Liang et al., 2015) [\[60\]](#)
- Faster RCNN (Ren et al., 2015) [\[61\]](#)
- Inside-Outside Net (Bell et al., 2015) [\[62\]](#)
- YOLO9000 (Redmon et al., 2016) [\[63\]](#)
- R-FCN (Dai et al., 2016) [\[64\]](#)
- Mask R-CNN (He et al., 2017) [\[65\]](#)
- Video Object segmentation (Caelles et al., 2017) [\[66\]](#)



## Visual Tracking

- Choi et al. (2017) [\[67\]](#)
- Yun et al. (2017) [\[68\]](#)
- Alahi et al. (2017) [\[69\]](#)

## Image Captioning



Retr.

1. Top view of the lights of a city at night, with a well-illuminated square in front of a church in the foreground;  
2. People on the stairs in front of an illuminated cathedral with two towers at night;

1. Tourists are sitting at a long table with beer bottles on it in a rather dark restaurant and are raising their bienglaeser;  
2. Tourists are sitting at a long table with a white table-cloth in a somewhat dark restaurant;

Gen.

A square with burning street lamps and a street in the foreground;

Tourists are sitting at a long table with a white table cloth and are eating;

- Mao et al. (2014) [\[70\]](#)
- Mao at al. (2015) [\[71\]](#)
- Kiros et al. (2015) [\[72\]](#)
- Donahue et al. (2015) [\[73\]](#)
- Vinyals et al. (2015) [\[74\]](#)
- Karpathy et al. (2015) [\[75\]](#)
- Fang et al. (2015) [\[76\]](#)
- Chen et al. (2015) [\[77\]](#)



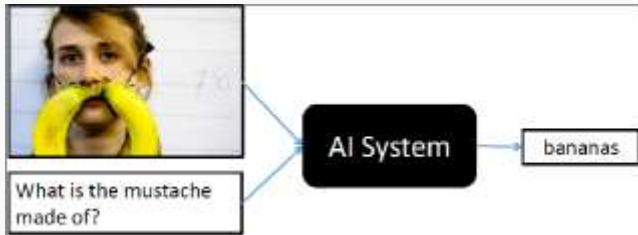
A group of young men playing a game of soccer



A man riding a wave on top of a surfboard.

## Video Captioning

- Donahue et al. (2014) [\[78\]](#)
- Venugopalan et al. (2014) [\[79\]](#)
- Pan et al. (2015) [\[80\]](#)
- Yao et al. (2015) [\[81\]](#)
- Rohrbach et al. (2015) [\[82\]](#)
- Zhu et al. (2015) [\[83\]](#)
- Cho et al. (2015) [\[34\]](#)



## Visual Question Answering

- Santoro et al. (2017) [\[84\]](#)
- Hu et al. (2017) [\[85\]](#)
- Johnson et al. (2017) [\[86\]](#)
- Ben-younes et al. (2017) [\[87\]](#)
- Malinowski et al. (2017) [\[88\]](#)
- Kazemi et al. (2016) [\[89\]](#)



## Video Question Answering

- Tapaswi et. al. 2016 [\[90\]](#)
- Zeng et. al. 2016 [\[91\]](#)
- Maharaj et. al. 2017 [\[92\]](#)
- Zhao et. al. 2017 [\[93\]](#)
- Yu Youngjae et. al. 2017 [\[94\]](#)
- Xue Hongyang et. al. 2017 [\[95\]](#)
- Mazaheri et. al. 2017 [\[96\]](#)



## Video Summarization

- Chheng 2007 [\[97\]](#)
- Ajmal 2012 [\[98\]](#)
- Zhang Ke 2016 [\[99\]](#)
- Zhong J 2017 [\[100\]](#)
- Panda 2017 [\[101\]](#)





## Generating Authentic Photos

Variational Autoencoders

(Kingma et. al., 2013) [\[102\]](#)

Generative Adversarial

Networks (Goodfellow et. al., 2014) [\[103\]](#)

Plug & Play generative nets  
(Nguyen et al., 2016) [\[104\]](#)

Progressive Growing of GANs  
(Karras et al., 2017) [\[105\]](#)

**Calls for Sanity (Interpretable, Fair, Responsible, Green AI)**

# The Paradox of Deep Learning

Why does deep learning work so well despite



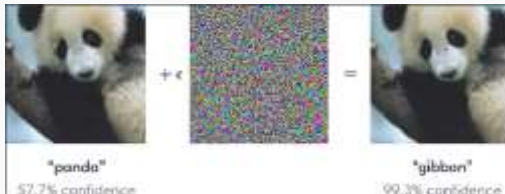
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\*<https://arxiv.org/pdf/1710.05468.pdf>

# The Paradox of Deep Learning

Why does deep learning work so well despite

- high capacity (susceptible to overfitting)



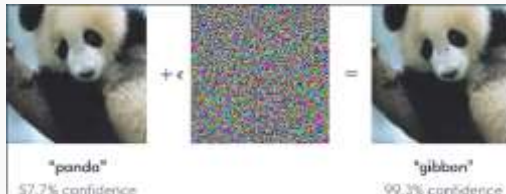
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Why does deep learning work so well despite

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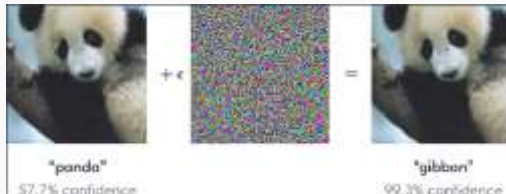
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Why does deep learning work so well despite

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- sharp minima (leading to overfitting)



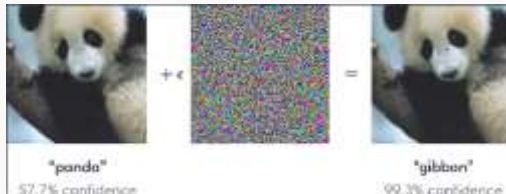
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- sharp minima (leading to overfitting)
- non-robustness (see figure)



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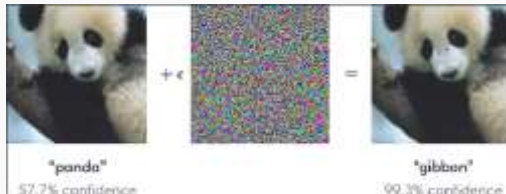
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No clear answers yet but ...



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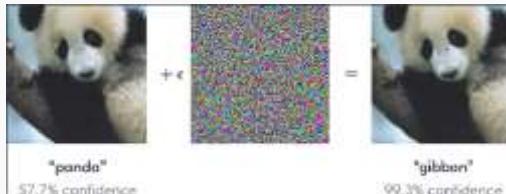
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No clear answers yet but ...

- Slowly but steadily there is increasing emphasis on explainability and theoretical justifications!\*



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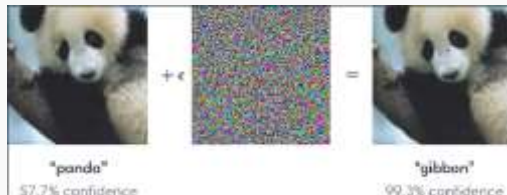
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No clear answers yet but ...

- Slowly but steadily there is increasing emphasis on explainability and theoretical justifications!\*
- Hopefully this will bring sanity to the proceedings !



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# Tell me why!

Workshop on Human Interpretability in  
Machine Learning

*We still do not know much about why DL models  
do what they do!*



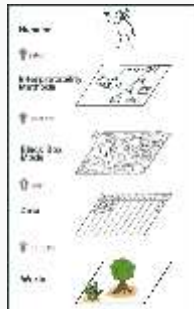
2016



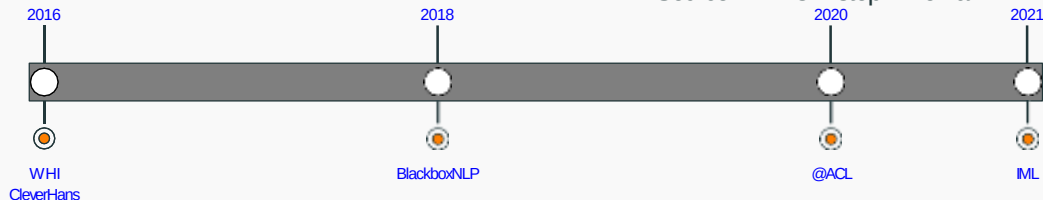
WHI

# Tell me why!

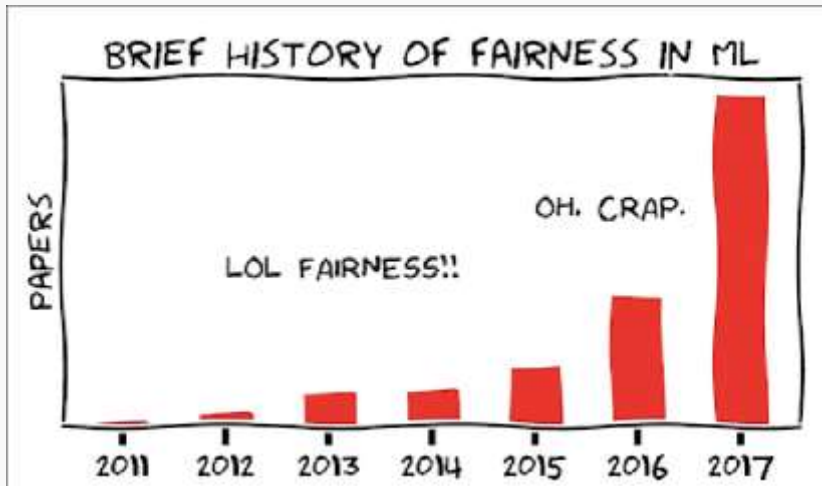
*Interpretable Machine Learning: A Guide for  
Making Black Box Models Explainable. —  
Christoph Molnar*



Source: IML: Christoph Molnar



## Be Fair and Responsible!



Source: <https://fairmlclass.github.io/> (Moritz Hardt)

## Be Fair and Responsible!

“There’s software used across the country to predict future criminals. And it’s biased against blacks.” - Propublica



Source:

<https://www.propublica.org/article/machine-bias->

2016



Machine Bias

# Be Fair and Responsible!

“Facial Recognition Is Accurate, if You’re a White Guy” - MIT Media

**Gender Shades audit, 2018**

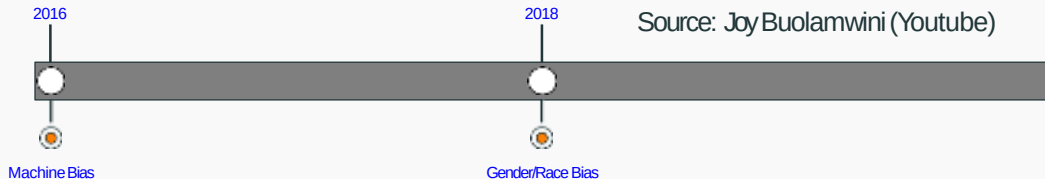
Accuracy in gender classification

|          | Lighter Male | Lighter Female | Darker Male | Darker Female | Lighter Male | Lighter Female | Darker Male | Darker Female |
|----------|--------------|----------------|-------------|---------------|--------------|----------------|-------------|---------------|
| Accuracy | 91.5%        | 91.5%          | 87.5%       | 87.5%         | 91.5%        | 91.5%          | 87.5%       | 87.5%         |
| Gender   | 91.5%        | 91.5%          | 87.5%       | 87.5%         | 91.5%        | 91.5%          | 87.5%       | 87.5%         |
| Gender   | 91.5%        | 91.5%          | 87.5%       | 87.5%         | 91.5%        | 91.5%          | 87.5%       | 87.5%         |

Source: Joy Buolamwini (Youtube)



Source: Joy Buolamwini (Youtube)



# Be Fair and Responsible!

Microsoft refuses to sell police its facial-recognition technology, following similar moves by Amazon and IBM

IBM says it is no longer working on face recognition because it's used for racial profiling



Microsoft won't sell police its facial-recognition technology, following similar moves by Amazon and IBM

2016



Machine Bias

2018



Gender/Race Bias  
Recognition Bias

2020

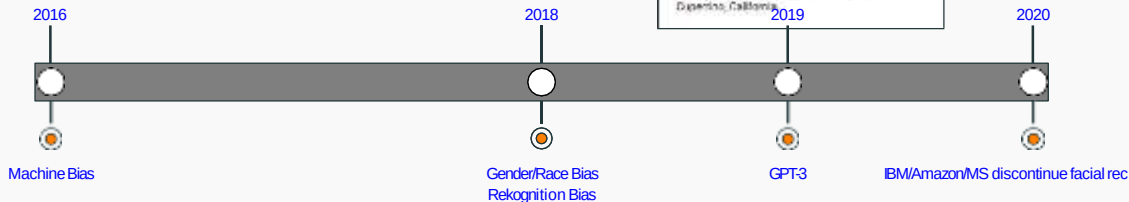
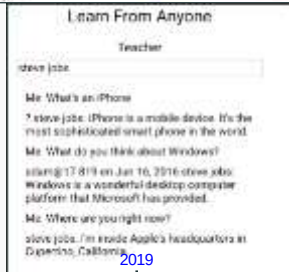


IBM/Amazon/MS discontinue facial rec



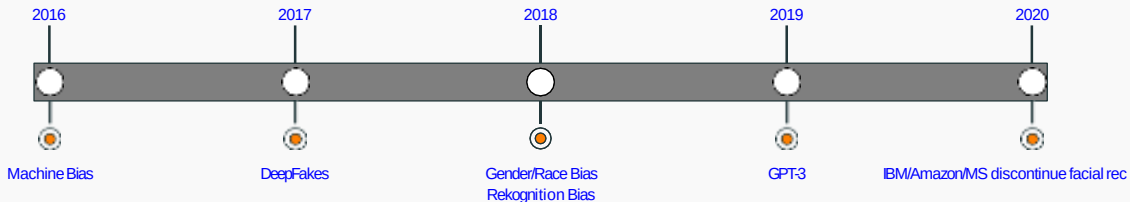
## Be Fair and Responsible!

“Due to our concerns about malicious applications of the technology, we are not releasing the trained model.” — *OpenAI*



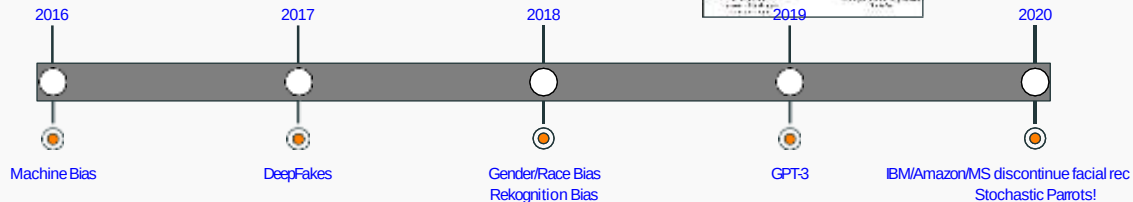
## Be Fair and Responsible!

What started off as an innocuous project for mimicking facial expressions has since lead to many apps and creation of fake videos for blackmailing, pronography and swaying elections!



# Be Fair and Responsible!

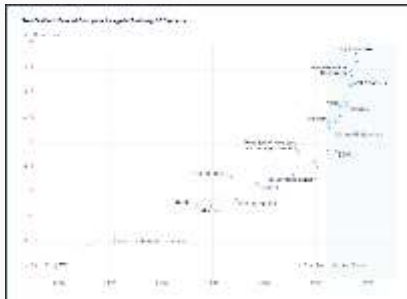
“Models are only as good as the data. Be responsible while curating data.” – *Bender et al.*



## Push for Green AI

The computations required for deep learning research have been doubling every few months, resulting in an estimated 300,000x increase from 2012 to 2018 – *AllenAI*

*Ironically, deep learning was inspired by the human brain, which is remarkably energy efficient.*



<https://openai.com/blog/ai-and-compute/>

2019

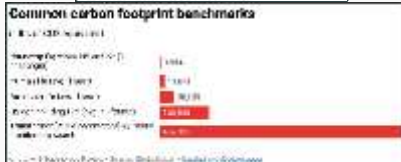


GreenAI

# Push for Green AI

Call for energy and policy considerations for Deep Learning

|                   | Global<br>emissions<br>per GPU | Per GPU<br>emissions<br>per day | Global<br>emissions<br>per day | Global<br>emissions<br>per year |
|-------------------|--------------------------------|---------------------------------|--------------------------------|---------------------------------|
| GPU 10000 x 10000 | 10000                          | 10000                           | 10000                          | 10000000                        |
| GPU 10000 x 10000 | 10000                          | 10000                           | 10000                          | 10000000                        |
| GPU 10000 x 10000 | 10000                          | 10000                           | 10000                          | 10000000                        |
| GPU 10000 x 10000 | 10000                          | 10000                           | 10000                          | 10000000                        |
| GPU 10000 x 10000 | 10000                          | 10000                           | 10000                          | 10000000                        |
| GPU 10000 x 10000 | 10000                          | 10000                           | 10000                          | 10000000                        |
| GPU 10000 x 10000 | 10000                          | 10000                           | 10000                          | 10000000                        |
| GPU 10000 x 10000 | 10000                          | 10000                           | 10000                          | 10000000                        |
| GPU 10000 x 10000 | 10000                          | 10000                           | 10000                          | 10000000                        |
| GPU 10000 x 10000 | 10000                          | 10000                           | 10000                          | 10000000                        |



2019



GreenAI  
Energy-Aware NLP

<https://github.com/ChristosChristofidis/awesome-deep-learning>

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